

Radiotherapy for head and neck cancer

Radiotherapy uses high-energy x-rays to destroy cancer cells, while doing as little harm as possible to normal cells.

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What is radiotherapy?

Radiotherapy is the use of high-energy x-rays to destroy cancer cells. It is an important treatment for [head and neck cancer](#).

Radiotherapy may be used on its own but is often given in combination with [chemotherapy](#). This is called chemoradiation.

Chemoradiation

Chemoradiation is often the main treatment for head and neck cancer. The chemotherapy helps the radiotherapy to work better.

It may be used:

- as your main treatment, if the cancer is locally advanced
- for cancers that cannot be removed with surgery
- when surgery could have serious effects on your speech or swallowing
- after surgery, to reduce the risk of the cancer coming back (adjuvant treatment)
- to treat cancers in harder-to-reach areas, such as the [nasopharynx](#) or throat ([oropharynx](#)).

[Chemoradiation](#) can cause severe side effects so it may not be suitable for everyone. If you are not well enough to cope with the side effects of chemoradiation, you may be given radiotherapy on its own.

Radiotherapy for early-stage cancers

Radiotherapy can be used on its own to treat cancers that are small and have not spread. It is often used to treat cancers in harder-to-reach areas. For example, it may be used for cancers in the back of [the mouth or throat](#). Radiotherapy can also be the main treatment when surgery could have a serious effect on your [speech](#) and [swallowing](#).

Radiotherapy for locally advanced cancers

If a cancer is bigger, or is affecting other tissues nearby (locally advanced cancer), radiotherapy is usually combined with other treatments. It may be given:

- after [surgery](#) (with or without chemotherapy) to destroy any remaining cancer cells and reduce the risk of cancer coming back
- in combination with [chemotherapy](#) (chemoradiation), without surgery
- in combination with the [targeted therapy](#) drug [cetuximab](#)
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to reduce symptoms (palliative radiotherapy).

Radiotherapy to control the cancer and reduce symptoms

If it is not possible to cure a cancer, treatment will aim to control the cancer and relieve symptoms. This is sometimes called palliative treatment.

You may have palliative radiotherapy to:

- help control the cancer for as long as possible
- stop bleeding from a tumour
- shrink a tumour that is causing swallowing or breathing difficulties
- relieve symptoms if the cancer has spread to other parts of the body, such as the [lungs](#) or [bones](#).

How radiotherapy is given

Radiotherapy is usually given from outside the body as [external-beam radiotherapy](#). A beam of x-rays is directed at the cancer from a large machine called a linear accelerator.

External-beam radiotherapy

External-beam radiotherapy is given in the hospital radiotherapy department. It can be used in different ways. It may be given:

- Monday to Friday, with a rest at the weekend – this is the most common method
- more than once a day
- every day, including at the weekend.

Treatment may take 4 to 7 weeks, depending on the type and size of the cancer. If you are having palliative radiotherapy, you may have a single treatment or 1 to 3 weeks of treatment.

Your cancer doctor (clinical oncologist), specialist nurse, or radiographer will discuss the treatment with you. They are part of [your radiotherapy team](#).

Intensity-modulated radiotherapy (IMRT)

[IMRT](#) is the most common type of external-beam radiotherapy used for head and neck cancer. It uses multiple beams of radiation that come from different directions, and shapes the beams to match the shape of the cancer. Shaping the beams lets the doctors give higher doses of treatment to the cancer and lower doses to the surrounding healthy tissue.

Research has found that, for some people, having IMRT rather than standard radiotherapy may reduce some [long-term side effects](#), such as a dry mouth. This is because the radiotherapy shapes the beams to avoid the salivary glands, which is where saliva (spit) is produced.

The treatment machines can also take pictures and scans. Some people may have frequent scans during their treatment. This is to make sure the treatment accurately targets the treatment area. Changes can be made if needed, for example if you lose weight during treatment.

Conformal radiotherapy (CRT)

Many types of external beam radiotherapy are conformal. This means the beams are specially shaped to fit the treatment area. [Conformal radiotherapy](#) can be used to treat many different types of cancer.

Proton beam therapy

[Proton beam therapy](#) uses proton radiation rather than x-rays to destroy cancer cells. Proton beams can be made to stop when they reach the area being treated. This is different to standard radiotherapy beams, which pass through the area and some healthy tissue around it. Proton beam radiotherapy is not widely available in the UK and is only suitable for a small number of people.

Planning your radiotherapy

To make sure your [radiotherapy](#) is as effective as possible, it has to be carefully planned. Planning makes sure the radiotherapy beams are aimed precisely at the cancer and cause the least possible damage to the surrounding healthy tissues. The treatment is carefully planned by a team of experts. This includes your cancer doctor and the radiographer who gives you the radiotherapy.

Planning is usually done during one visit to the radiotherapy department. It usually takes about 2 hours. You will meet the radiographers, who are experts in giving radiotherapy. They will answer any questions you have.

Radiotherapy mask

To help you lie in exactly the same position during radiotherapy, you wear a [plastic mesh mask](#). This is made individually for you. It holds your head and neck as still as possible. You have it on for up to about 15 minutes at a time. You can see and breathe normally while wearing it. It fits tightly but should not be uncomfortable. Most people get used to it quickly. It is sometimes called a mould, head shell or cast.

The radiographer and the technician who makes the mask will explain what is involved. But it is important to tell your radiographer or nurse if you find it difficult to wear the mask. They can suggest things that might help.

After the mask is made

When the mask is ready, your treatment can be planned. You will have a [CT scan](#) of the area to be treated. The radiographers take measurements to plan the treatment specifically for you. Some people also have an [MRI scan](#) as part of their radiotherapy planning.

The radiographer's measurements and the information from the scans are entered into the radiotherapy planning computer. This is used to plan your treatment precisely.

Sometimes, the radiographer draws marks on your mask. These help them to position you accurately each day before treatment. They may also make a very small permanent mark on your chest to help them position the mask.

Rarely, radiotherapy to the mouth or throat can cause swelling in tissues around the airways. This can make breathing difficult. If this is likely, your doctors arrange for you to have a small opening made in your windpipe before you have radiotherapy. This is called a [tracheostomy](#) (or stoma) and allows you to breathe comfortably. A tracheostomy in this situation is usually temporary. If you have any breathing difficulties during radiotherapy, contact your doctor straight away.

Having radiotherapy

Before each treatment session, the radiographer will position you on the treatment couch and carefully fit your mask. They will leave the room before the treatment starts. The treatment only takes a few minutes. There is a camera in the room, so the radiographer can see you from the next room. They can also hear you through an intercom. If you need help, you can raise your hand and the radiographers will return to the room.

External radiotherapy does not make you radioactive. It is safe for you to be with other people, including children and pregnant people, throughout your treatment.



Radiotherapy to the head and neck causes [side effects](#). Your [radiotherapy team](#) will tell you what to expect. They will explain how side effects can be controlled and what you can do to help to manage them.

We have more information about [side effects of radiotherapy for head and neck cancer](#).

Getting support

Macmillan is also here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#).
- [Visit our head and neck cancer forum](#) to connect with people who have been affected by head and neck cancer, share your experience, and ask questions.

About our information

References



Below is a sample of the sources used in our head and neck cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Machiels J.-P, Leemans C. R. et al. Squamous cell carcinoma of the oral cavity, larynx, oropharynx and hypopharynx. EHNS- ESMO-ESTRO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2020. Volume 31, Issue 11, Pages 1462-1475.

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, Dr Chris Alcock, Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help